Sumaiya Nazeen

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Education

Ph.D. in Computer Science
MIT, USA (September 2019)
Supervisor: Prof. Bonnie Berger

S.M. in Computer Science
MIT, USA (September 2014)
Supervisor: Prof. Bonnie Berger

B.Sc. Engg. in Comp. Sci. & Engg.
BUET, Bangladesh (February 2011)
Supervisor: Prof. M. Sohel Rahman

Postdoctoral Training

Postdoctoral Research Fellow
Area: Statistical Genetics and Neurogenomics
Institutions: Harvard Medical School & Brigham and Women's Hospital, USA
Years: November 2019 – Present
Mentors: Professors Shamil Sunyaev & Vikram Khurana

Postdoctoral Associate
Area: Computational biology
Institution: MIT, USA
Years: September 2019 – November 2019
Mentor: Prof. Bonnie Berger

Professional Experience & Affiliations

2022 – Present **Computational biology consultant**
Dacapo Brainscience, Inc., Cambridge, MA, USA
- Co-led the development of the core computational strategy and architecture of the company
- Actively supervise the computational team on developing proprietary computational strategies for target identification, patient stratification, and clinical trial design

2013 – Present **Remote associate researcher**
Broad Institute of MIT and Harvard, Cambridge, MA, USA

2024 – Present **Investigator**
Impact of Genomic Variation on Function (IGVF) Consortium, NHGRI, USA

2022 – Present **Statistical genetics research consultant**
MSA Coalition, Mission MSA, USA

2022 – 2023 **Genetics consultant**
Computational Genome Analysis Platform (CGAP), Harvard Medical School, Boston, MA, USA

2012 – 2019 **Graduate research assistant**
Computation and biology group, CSAIL, MIT, USA with Prof. Bonnie Berger

Spring 2017 **Graduate teaching assistant**
Departments of EECS and Mathematics, MIT, USA with Prof. Bonnie Berger

2011 – 2012 **Lecturer**
Department of Computer Science & Engineering, BUET, Dhaka, Bangladesh

Publications

1. **Nazeen S**, Wang X, Morrow AR, …, Khurana V & Sunyaev SR. (2026). NERINE reveals rare variant associations in gene networks in multiple phenotypes and implicates an *SNCA-PRL-LRRK2* subnetwork in Parkinson's disease. *Cell Genomics* 2026, 6 (7). (featured on the journal cover).

2. Kar P, Moldovan M, Guez J, **Nazeen S**, Goodrich JK, Karani T, Samocha KE, Karczewski K, Koch E, Seplyarskiy V & Sunyaev SR. (2026). Inference of elevated mutation rates and variant effects using 700k exomes. *bioRxiv* 2026. (under review at *Science*)

3. Mitchel J, **Nazeen S**, Wang X, …, Kharchenko PV & Sunyaev SR. (2025). A single-cell genetic colocalization test improves power and resolves disease-mediating cell types. *bioRxiv* 2025. (under 2nd stage review at *Nature Genetics*)

4. Noori A, Polonuer J, Meyer K, Budnik B, Morton S, Wang X, **Nazeen S**, He Y, Arango I, Vittor L, Woodworth M, Krolewski R, Li M, Liu N, Kamath T, Macosko E, Rodriques S, Dagan N, Das S, Church G, Tam J, Khurana V & Zitnik M. Graph AI generates neurological hypotheses validated in molecular, organoid, and clinical systems. *arXiv* 2025 (under review at *Nature)*

5. Cox LM*, Korecka JA*, Beauchamp LC, Compagnoni GM, Acera MA, Murueta-Goyena A, Silva P, Gemos A, Sheinkopf Z, Li S, Rountree R, Schwerdtfeger L, Barrasa MI, **Nazeen S**, Fan B, …, Weiner HL & Khurana V. Host-microbe interactions shape brain vulnerability in familial Parkinson's disease and dementia. 2026. (under review at *Nature*)

6. Murdock DR, Guo D, Guan P, Bermudez F, DePaolo J, Cabot J, **Nazeen S**, Nasir H, Gupta RM, Jha A, Elefteriades J, McGivern B, Anderson SK, Jones C, Damrauer S, Chen H & Milewicz DM. Gene-based, Rare Variant Burden Analyses of Biobank Data Reveal Novel Genes Conferring a High Risk for Thoracic Aortic Disease. *medRxiv* 2026. (under review at *Circulation: Genomic and Precision Medicine*)

7. **Nazeen S***, Wang X*, Zielinski D*, Lam I, Hallacli E, Xu P, ..., Sunyaev SR & Khurana V. (2024). Deep sequencing of proteotoxicity modifier genes uncovers a *Presenilin-2/beta-amyloid-actin* genetic risk module shared among alpha-synucleinopathies. *bioRxiv* 2024. (* denotes co-first authors)

8. Bendapudi P, **Nazeen S**, Ryu J, Söylemez O, …, Maas R, Sunyaev SR & Losman JA. Low-frequency inherited complement receptor variants are associated with purpura fulminans. *Blood* 2024, 143 (11): 1032–1044.

9. Connally N, **Nazeen S**, Lee D, Shi H, Stamatoyannopoulos J, Chun S, Cotsapas C, Cassa C, & Sunyaev S. The missing link between genetic association and regulatory function. *Elife* 2023. 11: e74970.

10. Hallacli E, Kayatekin C, **Nazeen S**, Wang X, Sheinkopf Z, Sathyakumar S, Sarkar S, Jiang X, Dong X, DiMaio R, Wang W, Keeney M, Felsky D, Sandoe J, Vahdatshoar A, Udeshi N, Mani DR, Carr S, Lindquist S, Jager P, Myers C, Bartel D, Greenamyre T, Feany M, Sunyaev SR, Chung CY & Khurana V. The Parkinson's disease protein alpha-synuclein is a direct modulator of RNA processing-bodies and mRNA stability. *Cell* 2022. 185(12): 2035-2056.e33.

11. **Nazeen S**, Yu YW & Berger B. Carnelian uncovers hidden functional patterns across diverse study populations from whole metagenome sequencing reads. *Genome Biology*. 2020: 21(1): 47.

12. **Nazeen S*,** Palmer NP*, Berger B & Kohane IS. Integrative analysis of genetic datasets reveals a shared innate immune component in autism spectrum disorder and its co-morbidities. *Genome Biology*. 2016; 17(1): 228. (* denotes co-first authors)

13. **Nazeen S***, Rahman MS & Reaz R*. Indeterminate String Inference Algorithms. *Journal of Discrete Algorithms*. 2012; 10: 23-34. (* denotes co-first authors; author names are in alphabetical order)

14. Moosa TM, **Nazeen S***, Rahman SM & Reaz R*. Inferring Strings from Cover Arrays. *Discrete Mathematics, Algorithms and Applications*. 2013; 5(02): 1360005. (* denotes co-first authors; author names are in alphabetical order)

15. Moosa TM, **Nazeen S***, Rahman MS & Reaz R*. Linear Time Inference of Strings from Cover Arrays using a Binary Alphabet. Md. S. Rahman and S.-i. Nakano (eds.). *Proceedings of WALCOM 2012. LNCS,* vol. 7157, pp. 160-172. Springer, Heidelberg. (* denotes co-first authors; author names are in alphabetical order; peer-reviewed)

Manuscripts in preparation

1. **Nazeen S***, Chiang C*, …, Cassa CS, Khurana V & Sunyaev SR. Detection of genetic subtypes of complex diseases by eigenanalysis. (2026). (* denotes co-first authors)

2. Ding K*, **Nazeen S***, Li MM, …, Zitnik M, Luo Y & Khurana V. TransposeNet-2.0: Cross-species structure- and context-aware learning enhances functional consistency in human protein interaction networks. (2026). (* denotes co-first authors)

3. **Nazeen S***, El-Kodsi D*, Rodriguez D*, ..., Khurana V. Harvard Biomarkers Study 2.0: A Longitudinal Multi-modal Platform for Biomarker Discovery in Parkinson's Disease and Related Disorders. (2026). (* denotes co-first authors)

4. **Nazeen S***, Nasir H*, Patnaik PK, Wang X, Morrow A, Ariet MAS, Dirie S, Rodriguez D, El-Kodsi D, Barrasa MI, Khurana V. Convergent Human Genetic and Functional Genomic Evidence Implicates *DOLPP1* in Parkinson's Disease. (2026). (* denotes co-first authors)

5. Wang X*, Patnaik, P*, **Nazeen S**, …, Sunyaev SR & Khurana V. Scalable HiPSC-based model for characterizing genetic modifiers of synucleinopathies. (2026).

6. Nguyen TTL, **Nazeen S**, …, Sunyaev SR, Tewhey R & Reilly SK. Dissecting regulatory landscape of human complex traits with perturb-seq. (2026). *ENCODE consortium companion paper*.

7. Dong A*, Li MM*, Ding K, **Nazeen S**, Noori A, Huang Y, Gao S, Ritter D, Afroz J, Henderson ABH, Studer L, Luo Y, Khurana V, Parmigiani G & Zitnik M. A cell type resolved protein representation model for central nervous system. (2026).

Thesis

1. **Nazeen S**. Computational methods for functional interpretation of diverse omics data. Ph.D. Thesis (2019). Massachusetts Institute of Technology.

2. **Nazeen S**. Integrative analysis of heterogeneous genomic datasets to discover genetic etiology of autism spectrum disorders. S.M. Thesis (2014). Massachusetts Institute of Technology.

3. **Nazeen S** and Reaz R. Indeterminate string inference algorithms. B.Sc. Engg. Thesis (2011). Bangladesh University of Engineering and Technology.

Conference presentations & posters

2026 **Convergent Human Genetic and Functional Genomic Evidence Implicates *DOLPP1* in Parkinson's Disease**
Poster presentation at **PQG 2026: 19th Annual Conference of the Program in Quantitative Genomics**, Boston, MA, USA

2025 **Uncovering novel disease mechanisms through network-aware rare variant testing with NERINE**
Poster presentation at **ASHG 2025: American Society of Human Genetics annual meeting** (Statistical Genetics and Genetic Epidemiology track), Boston, MA, USA

2025 **Latent structure modeling of genetic heterogeneity in complex diseases**
Poster presentation at the **Gordon Research Conference: Human Genetics and Genomics**, 2025, University of Southern Maine, ME, USA

2024 **NERINE: Network-based association test for rare variants in complex traits and diseases**
Selected abstract for **full-length talk** at **ESHG 2024: European Human Genetics Conference** (Machine Learning and Statistics session), Berlin, Germany

2019 **Uncovering hidden functional patterns across diverse study populations from whole metagenome sequencing reads with Carnelian**
Highlights talk at the **Workshop on Microbiomics, Metagenomics, and Metabolomics (ACM-BCB 2019)**, Niagara Falls, NY, USA

2018 **Carnelian: enhanced functional profiling of metagenomic reads**
Selected abstract for **full-length talk** at the **17th European Conference on Computational Biology (ECCB 2018)** (Applications Track), Athens, Greece

2015 **LAPRAS: An integrative model incorporating heterogeneous datasets to discover genetic etiology of autism spectrum disorder**
Selected abstract for short **oral presentation** and poster presentation at the **23rd Annual International Conference on Intelligent Systems for Molecular Biology and 14th European Conference on Computational Biology (ISMB/ECCB 2015)**, Dublin, Ireland

2014 **Integrative Analysis of Multiple Gene Expression Studies Reveals Genes and Pathways involved in Alteration of Steroidogenesis in Polycystic Ovary Syndrome (PCOS)**
Poster presentation at the **13th European Conference on Computational Biology (ECCB 2014)**, Strasbourg, France

2012 **Linear Time Inference of Strings from Cover Arrays using a Binary Alphabet**
Selected abstract for **full-length talk** at the **6th International Workshop on Algorithm and Computation (WALCOM 2012)**, Dhaka, Bangladesh

Invited talks

From Networks to Subtypes: Statistical Frameworks for Mechanistic Insights into Complex Disease Genetics

2026 ACM Seminar Series, BUET ACM Chapter, Bangladesh University of Engineering and
2025 Technology
Bioinformatics Seminar Series, Massachusetts Institute of Technology

Network-aware rare variant association in complex diseases

2026 MPG Primer Series, Broad Institute of MIT and Harvard (video)
2024 AD/PD Seminar Series, Division of Movement Disorders, Department of Neurology, Brigham and Women's Hospital
2024 Genetics Research in Progress (GRIP) Meeting, Division of Genetics, Brigham and Women's Hospital
2024 Medical and Population Genetics (MPG) Research Meeting, Broad Institute of MIT and Harvard
2023 Statistical Genetics Meeting, Broad Institute of MIT and Harvard

Computational Biology in the 21st Century: Untangling Health and Disease

2023 AABEA Great Lakes Seminar Series, MI

Network-based genomics approaches for rare variants in synucleinopathies

2023 Aligning Science Across Parkinson's (ASAP) GWAS meeting
2021 International Parkinson's Disease Genomics Consortium (IPDGC) Meeting, The National Institutes of Health (NIH)

Understanding the genetic foundations of synucleinopathies with statistical and ML approaches

2022 Machine Learning for Healthcare (ML4H) Seminar Series, Broad Institute of MIT and Harvard

Functional Interpretation of Omics Data for Health and Disease

2020 ACM Seminar Series, BUET ACM Chapter, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh

Uncovering hidden functional patterns across diverse study populations from whole metagenome sequencing reads with Carnelian

2018 Center for Microbiome Informatics and Therapeutics, Broad Institute of MIT and Harvard
2018 Frontiers in Mucosal Immunology (FIMI) symposium, Harvard Medical School, Boston, MA, USA

Honors & Awards

2026	Alumni Excellence Award	Holy Cross College, Dhaka, Bangladesh	Awarded for excellence in research and academia at the 75 years jubilee celebration of the college
2022	Sudarsky Scholar Award	Movement Disorders Division, Department of Neurology, Brigham and Women's Hospital	Awarded for promising and innovative independent project
2015	Ludwig Center Graduate Fellowship	The Koch Institute of Integrative Cancer Research, and MIT School of Science	Excellence in Ph.D. Research

2012	International Fulbright Science & Technology Award	United States Department of State's Bureau of Educational and Cultural Affairs	Awarded to the most promising future researchers and leaders for pursuing Ph.D.
2011	Chancellor's Award	Bangladesh University of Engineering and Technology	Highest CGPA at the bachelor's level
2011	Prime Minister's Gold Medal	University Grants Commission, Bangladesh	Excellence in undergraduate studies
2006 – 2010	Dean's List Award	Bangladesh University of Engineering and Technology	Highest CGPA in every semester of undergraduate studies

Teaching

Spring 2017, MIT	Spring 2012, BUET	Fall 2011, BUET	Spring 2011, BUET
18.418: Advanced Topics in Computational Molecular Biology	CSE 409, CSE410: Computer Graphics CSE 462: Algorithm Engineering CSE 464: Computational Geometry CSE 472: Machine Learning	CSE 201, CSE 202: Object-Oriented Programming Language CSE 310: Compilers CSE 410: Computer Graphics CSE 464: Computational Geometry CSE 472: Machine Learning	CSE 100S: Introduction to Computer Systems CSE 106: Structured Programming Language CSE 214: Assembly Language Programming

Professional Services

Ad hoc reviewer	*Nature Communications, Bioinformatics, JAMIA, Clinical and Translational Medicine*, RECOMB, ISMB, ACM-BCB, *Nature (co-reviewer), NEJM (co-reviewer), Annals of Neurology (co-reviewer)*
Program committee member	RECOMB 2027
Organizing committee member	RECOMB 2017
Expert panelist	South Asian Regional Conference on Higher Education (SARCHE) 2026

Mentoring

2026 – Present	Sadat Hossain, Graduate Research Assistant, Bangladesh University of Engineering & Technology (Master's thesis)
	Sojib Zaman, Graduate Research Assistant, Bangladesh University of Engineering & Technology (Master's thesis)
	Anik Saha, Undergraduate Research Assistant, Bangladesh University of Engineering & Technology (Senior thesis)
	Yung-Chen Hsieh, Undergraduate Research Assistant, Stanford University
2026 summer	Leander Goessling, Undergraduate Research Assistant, University of California, San Diego
	Lucia Chen, Research Summer Institute (RSI), MIT
	Aishwarya Srinivas, Summer Research Assistant, HMS
2025 summer	Evrim Sude Con, Undergraduate Research Assistant, Harvard University
2025 summer	Yung-Chen Hsieh, Research Summer Institute (RSI), MIT (now at Stanford University)
	Ariv Vaidya, Summer Research Assistant, BWH, HMS (now at Johns Hopkins University)
2024 – Present	Habib Nasir, Graduate Research Assistant, BWH, HMS
2024 summer	Angela Zhan, Research Summer Institute (RSI), MIT (now at Stanford University)

2023 – 2024	Autumn R. Morrow, Graduate Research Assistant, BWH, HMS (now at Vanderbilt University)
2021 – 2023	Sarah Jensen, Summer Research Assistant, BWH (now at Tufts University)
Fall 2011	Six undergraduate student projects in the Object-Oriented Programming Language course, BUET
Spring 2011	Ten undergraduate student projects in the Structured Programming Language course, BUET

Press

2026	**New Tool Helps Uncover Rare Genetic Mutations in Common Diseases, Including Parkinson's** https://hms.harvard.edu/news/new-tool-helps-uncover-rare-genetic-mutations-common-diseases-including-parkinsons
2023	**Featured on the Faces of Cell** https://x.com/CellCellPress/status/1613951138432258083
2023	**Building momentum against Parkinson's** https://news.harvard.edu/gazette/story/2022/09/inching-parkinsons-into-the-light/

References (available upon request)